

Predictors of outcome of anterior temporal lobectomy for intractable epilepsy

A multivariate study

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Article abstract—*Objective:* To identify presurgical and postsurgical factors that are independently predictive of the outcome of anterior temporal lobectomy (ATL) for intractable epilepsy. *Background:* There have been reports of prognostic factors in epilepsy surgery, but little is known about factors that independently predict outcome of ATL. *Methods:* We studied 175 consecutive ATL patients who had at least 2 years of postsurgical follow-up. Significant factors on univariate analyses were subjected to stepwise logistic regression analysis. *Results:* On univariate analyses, two presurgical conditions were significantly associated with excellent seizure control at last follow-up: (1) unilateral hippocampal formation atrophy as detected on MRI and (2) all scalp interictal epileptiform discharges concordant with the location of ictal onset ($p < 0.05$). Three postsurgical factors that occurred during the first year were associated with excellent seizure outcome: the absence of interictal epileptiform discharges at 3 months, complete seizure control, and having only nondisabling seizures for those who did not become seizure free. Logistic regression analysis revealed the following to be independently predictive of excellent seizure control: MRI-detected unilateral hippocampal formation atrophy, concordant interictal epileptiform discharges, complete seizure control during the first postsurgical year, and having only nondisabling seizures during the first postsurgical year for those who did not become seizure free. *Conclusions:* Presurgical identification of unilateral hippocampal formation atrophy, or of interictal epileptiform discharges that are all concordant with the location of ictal onset, predict excellent outcome of ATL. However, the probability of excellent outcome is highest (94%) when both factors are present.

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Anterior temporal lobectomy (ATL) remains the most effective and widely performed type of surgery for intractable epilepsy.¹ Its success rate has improved in the last decade. A multicenter survey showed that approximately two-thirds of patients become seizure free after the surgery.¹ Nonetheless, about one-third of patients continue to experience seizures after ATL. Hence, determining prognostic factors becomes important in identifying the ideal surgical candidate, in counseling patients about the risks and benefits of ATL, and in promoting the understanding of the phenomenon of seizure intractability.

Several studies have reported different factors that are associated with a favorable outcome of ATL. The first study to systematically evaluate temporal lobectomy patients and to apply statistical analysis was conducted almost 30 years ago by Bengzon et al.² The study found a greater number of prognostic factors (19) than nonprognostic factors (16). The study was not of a consecutive series of patients and was clearly of a biased population. Subjects were selected to represent the extremes of postsurgical

outcome—patients with the best possible outcome contrasted with patients with the worst outcome. Modern techniques of localizing the surgical focus, such as video-EEG monitoring and high-resolution MRI,³ were not evaluated. The authors also suggested that it would have been important to apply multivariate analysis, which at that time was just emerging as a statistical method. Since then, there have been only a few published reports^{4–7} that used multivariate analysis, and all except one⁶ consisted of a mixture of patients who had temporal or extra-temporal surgery. The exception, however, consisted of patients who had ATL before the advent of the use of MRI for presurgical evaluation. Moreover, one of the other studies was restricted to a group of patients whose EEG was recorded with intracranial electrodes,⁵ whereas another included patients who had repeat operations after the initial surgery failed to control seizures.⁴ Two of the three studies limited the assessment of outcome in all patients to only 2 postoperative years.^{4,7}

We report the results of a study with multivariate

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analysis to identify factors that are independently predictive of ATL outcome in a group of consecutive patients with intractable epilepsy.

Methods. *Description of cohort and presurgical evaluation.* Between January 1, 1988, and June 30, 1991, 190 patients underwent first-time ATL with amygdalohippocampectomy at the Mayo Clinic for control of medically intractable complex partial epilepsy. Patients with extratemporal surgery were not included. Seizure control was the main objective of the surgery. The cohort in this study was composed of 175 consecutive patients who had 2 or more years of postoperative follow-up (mean 3.6 years, range 2 to 5.7). (Four patients with 2 or more years of follow-up had insufficient data for study.) A multidisciplinary presurgical evaluation was performed, that included a complete medical and neurologic history and examination, brain MRI with temporal lobe volume protocol,⁸ wake and sleep outpatient EEG, ophthalmologic examination including visual field test, neuropsychological evaluation, video-EEG recording of seizures, cerebral angiograms, and intracarotid amobarbital test. The surgical focus was localized by the clinical, neuroimaging, and electrophysiologic findings. Ictal video-EEG recording was used to confirm the location of ictal onset. When presurgical evaluation yielded findings that were indeterminate, or when results of the presurgical tests were conflicting, electrodes were implanted in the temporal lobes, and in other extratemporal regions when necessary, for recording of intracranial ictal activity. ATL was considered and offered as a treatment option when presurgical evaluation localized the source of habitual seizures to one temporal lobe. In patients with multiple seizure foci on ictal recording, ATL was considered only when habitual seizures could be demonstrated to originate from one temporal lobe and to account for at least 90% of the recorded seizures, and when the rest of the presurgical evaluation showed no conflicting results.

Operative procedure. All patients underwent the same technique of resecting the lateral temporal cortex and the mesial temporal structures, which included the amygdala, the hippocampus, and the parahippocampal gyrus. The procedure was performed under general anesthesia. The posterior boundary hippocampal resection was at its posterior medial turn at the minimum. Maximal lateral resection was always attempted, but the resection was guided by the side of language dominance, the preresection electrocorticography findings,⁹ and the presence of any surgically restricting vascular structures such as the vein of Labbé.

Data collection. Patient charts were reviewed to collect information on demographics, epilepsy history, and postoperative seizure frequency. Unless a patient's condition required otherwise, routine postoperative follow-up visits were conducted on the first postoperative day and at 3 months, 1 year, and each year thereafter. At each visit, information collected included seizure frequency, medication intake and compliance, and the effect of seizures on daily activities, driving, and work. To minimize the lack of contribution to outcome determination from patients who did not return for clinic visits, latest information was obtained by using an intensive program of data collection. This program was approved by our Institutional Review

Board for our Survey Research Center to send out correspondence in January of 1993 after current addresses were verified or determined. Response rate was 72%. The Survey Research Center subsequently successfully contacted by phone 66% of the remaining patients.

The protocol of postoperative evaluation also included CT of the head on the first day, MRI of the head and neuropsychological evaluation at 3 months, and wake and sleep EEG during the first week, at 3 months, and at 1 year. The presence and location of scalp interictal epileptiform discharges recorded before surgery were determined from both outpatient routine EEGs and from prolonged video-EEG recordings. Outpatient EEGs were recorded on 20 channels, whereas prolonged EEG monitoring consisted of 32 channels of recording, including subtemporal chains of electrodes. During EEG monitoring, interictal activities were sampled for 20 seconds every half hour. Sphenoidal electrodes and computer programs for detecting spikes were not used.

Assessment of outcome. The Seizure Frequency Scoring System was used to assess the outcome of each patient in this cohort. This system was originally introduced by Engel et al.¹ We have recently demonstrated the practicality of using this system in following the seizure outcome of our patients throughout their entire postsurgical course.¹⁰ Patients with a seizure frequency score of 0 to 4 at last follow-up were considered to have had an excellent outcome because they were either seizure free or had only nondisabling seizures. Seizures were considered nondisabling when their occurrence did not interfere with ability to work or drive. We have previously validated this criterion for excellent outcome by demonstrating that it is associated with the best work outcome, with ability to drive, and with improvement in independent living activities after surgery.¹¹

Statistical analysis. The chi-square test was used for univariate analysis of the following categorical variables: epilepsy etiology (symptomatic versus cryptogenic), febrile seizure history, unilateral hippocampal formation atrophy on MRI, neuroimaging lesions other than hippocampal formation atrophy, concordance of interictal epileptiform discharges with the location of ictal onset, postresection spikes on electrocorticogram, interictal epileptiform discharges in the first postoperative week, interictal epileptiform discharges at 3 months after surgery, and excellent seizure control in the first postsurgical year (seizure frequency score of 0 to 4). The rank sum test was used to evaluate the following continuous variables: age at seizure onset, age at surgery, duration of epilepsy history before surgery, and length of follow-up. Multivariate analysis using stepwise logistic regression (stepping up) was performed on variables that were significant ($p < 0.05$) in univariate analysis. All subjects were included in the analysis. Missing data were handled by creating two variables. The first variable was coded with the actual observed value for subjects without missing values and with 0 for those with missing values. The second variable was coded as 0 if the information was present and as 1 if absent. In this way, we are able to include all subjects in the analysis and also estimate any associations resulting from the missing values.

Two measures were used to evaluate the prognostic value of the degree of seizure control during the first post-

Table 1 Characteristics of the 175 patients in the cohort

	n (%)	Median	Range
Male	77 (44.0)		
Female	98 (56.0)		
Handedness			
Right	151 (86.3)		
Left	20 (11.4)		
Ambidextrous	4 (2.3)		
Epilepsy types according to cause			
Symptomatic	105 (60.0)		
Cryptogenic	70 (40.0)		
Congenital CNS abnormalities	3 (1.7)		
History of perinatal distress*	5 (2.9)		
History of neonatal seizures	0		
History of febrile seizures	57 (32.6)		
History of encephalitis	11 (6.3)		
History of meningitis	8 (4.6)		
Subdural empyema	2 (1.1)		
Major brain trauma†	19 (10.9)		
History of status epilepticus	12 (6.9)		
Stroke	3 (1.7)		
Family history of seizures	4 (2.3)		
One or more seizures per week‡	135 (77.1)		
Previous acute (provoked) seizures	5 (2.9)		
Age at first unprovoked seizure, y		8.0	1–62
Age at surgery, y		31.0	7–86
Presurgical duration of epilepsy, y		19.0	<1–81

* For example, any of the following: Apgar score <5, intubation performed, or confinement in incubator or in intensive care unit.

† Resulting in loss of consciousness or in intracranial hemorrhage or contusion.

‡ During the year before surgery.

operative year: being seizure free and having only nondisabling seizures when not seizure free. Two variables were created for the analysis of these measures. The first variable was coded as 1 if seizure free and as 0 if not seizure free. The second variable was coded as 1 if seizures were nondisabling and as 0 if seizures were disabling.

Results. Table 1 shows the characteristics of patients in our cohort, including demographics, presurgical history and factors of epilepsy, presurgical seizure frequency, and age at surgery.

The two most prominent risk factors for the development of epilepsy were intracranial infection (12%) and major head trauma (approximately 11%). Approximately three-fourths of the cohort were experiencing at least one seizure per week before surgery, and approximately 7% had an episode of status epilepticus. Most patients have had a long history of epilepsy before surgery, with a median duration of 19 years.

Table 2 shows the results of the presurgical evaluation and the postsurgical outcome. Hippocampal formation at-

rophy is the most common, potentially epileptogenic lesion detected by MRI. A little less than one-third of the entire cohort had other lesions. Scalp EEG recorded unilateral interictal epileptiform discharges in almost three-fourths of all patients. Only a small proportion (1.1%) of patients had no interictal epileptiform discharges on scalp EEG, and the rest had bilateral independent temporal or concomitant extratemporal discharges. An average of four to five seizures were recorded in each patient. Approximately three-fourths of all patients had an excellent outcome at last postsurgical follow-up, being seizure free or having only nondisabling seizures.

On univariate analyses, two presurgical conditions were significantly associated with excellent seizure control at last follow-up: (1) unilateral hippocampal formation atrophy as detected by MRI and (2) all interictal epileptiform discharges concordant with the location of ictal onset (table 3). Presurgical factors that were not found to be significant were age at unprovoked seizure onset, duration of epilepsy history, epilepsy etiology, febrile seizure history, presence of brain lesions other than hippocampal formation atrophy, and age at surgery. Three postsurgical factors that occurred during the first year were associated with excellent seizure outcome: the absence of interictal epileptiform discharges at 3 months, complete seizure control, and having only nondisabling seizures for those who did not become seizure free. Other postsurgical factors

Table 2 Results of presurgical evaluation and postsurgical outcome in 175 anterior temporal lobectomy patients

Presurgical evaluation results	n (%)
MRI hippocampal formation atrophy*	77/143 (53.8)
Unilateral	74 (51.7)
Bitemporal	3 (2.1)
Other lesions on neuroimaging†	47/175 (26.8)
Temporal	30 (17.0)
Extratemporal	12 (6.9)
Temporal and extratemporal	5 (2.9)
Interictal epileptiform discharges (scalp)	173/175 (98.8)
Unilateral temporal	125 (71.4)
Bitemporal	44 (25.1)
Concomitant extratemporal	4 (2.3)
Intracranial electrodes implanted	11/175 (6.3)
Side of surgery	
Right	68 (38.9)
Left	107 (61.1)
Postresection electrocorticography spikes‡	54/171 (31.6)
Epileptiform discharge at 3 mo post surgery§	48/147 (32.7)
Major surgical complications¶	2/175 (1.1)
Excellent outcome	134/175 (77.0)

* Hippocampal volume assessment not performed in 32 patients; 15 of the 32 had sufficient studies for detection of other lesions.

† For example, cerebral vascular malformations and tumors.

‡ Electrocorticography not done in four patients.

§ Three-month postsurgical EEG not done in 28 patients.

¶ Moderate to severe hemiparesis and dysphasia.

|| Seizure-free or with only nondisabling seizures (mostly auras).

Table 3 Univariate analysis of factors associated with postsurgical outcome in 175 anterior temporal lobectomy patients

	Excellent (n = 134)	Nonexcellent (n = 41)	p Value
Presurgical factors			
Age at unprovoked seizure onset, y*	11.3 ± 10.3	12.5 ± 10.4	0.465
Symptomatic epilepsy etiology	42.5%	31.7%	0.215
Duration of epilepsy history, y*	19.7 ± 12.1	20.7 ± 11.6	0.566
History of febrile seizure	33.6%	29.3%	0.606
Unilateral hippocampal formation atrophy on MRI	47.0%	26.8%	0.016
Other lesions on neuroimaging	19.4%	9.8%	0.152
Concordant interictal epileptiform discharge†	73.9%	43.9%	<0.001
Age at surgery, y*	31.0 ± 11.4	33.2 ± 11.6	0.196
Postsurgical factors			
No epileptiform discharge on corticogram	69.4%	58.5%	0.118
No epileptiform discharge at 1 week	59.7%	51.2%	0.163
No epileptiform discharge at 3 months	62.7%	36.6%	0.005
Seizure free during first year	81.3%	17.1%	<0.001
Only nondisabling seizures during first year‡	10.4%	4.9%	<0.001
Length of follow-up, y*	3.6 ± 1.1	3.7 ± 1.0	0.705

* Mean ± SD.

† Scalp interictal epileptiform discharges concordant with ictal onset.

‡ For those not seizure free.

were not significant: epileptiform discharges on postresection electrocorticogram, interictal epileptiform discharges at 1 week, and the length of follow-up.

Logistic regression analysis revealed the following to be independently predictive of excellent seizure control: MRI-detected unilateral hippocampal formation atrophy ($p = 0.024$; odds ratio (OR) = 3.7 [95% CI = 1.2 to 11.7]); concordant interictal epileptiform discharges ($p = 0.019$; OR = 3.5 [95% CI = 1.2 to 9.7]); complete seizure control during the first postsurgical year ($p < 0.001$; OR = 54.9 [95% CI = 17.3 to 174.4]); and having only nondisabling seizures during the first postsurgical year for those who did not become seizure free ($p < 0.001$; OR = 22.7 [95% CI = 3.9 to 131.8]).

Overall, 85% (63/74) of the patients with unilateral hippocampal atrophy versus 63% (34/54) of those without it had excellent outcome ($p = 0.016$). We also determined and compared the absolute rates of achieving excellent outcome when either independently predictive presurgical factor was present, when both were present, and when both were absent. The figure shows that, in patients with potentially epileptogenic abnormalities on MRI, concordance of all interictal epileptiform discharges with location of ictal onset was significantly associated with a high probability of excellent postsurgical outcome. Ninety-four percent of patients (95% CI = 84.6 to 98.8) had excellent outcome when all interictal epileptiform discharges were concordant and when unilateral hippocampal formation atrophy was present, whereas only 60% (95% CI = 36.1 to 80.9) had excellent outcome when hippocampal formation atrophy alone was present ($p = 0.001$). The finding was similar in patients with MRI lesions other than hippocampal formation atrophy (95% [95% CI = 76.2 to 99.9] excellent outcome in patients with all interictal epileptiform

discharges concordant versus 67% [95% CI = 29.9 to 92.5] in those without; $p = 0.035$) but not in patients with MRI negative for hippocampal formation atrophy or for other potentially epileptogenic lesions (66% [95% CI = 46.8 to 81.4] excellent outcome in patients with all interictal epileptiform discharges concordant versus 59% [95% CI = 36.4 to 79.3] in those without; $p = 0.625$). Thus, concordance of all interictal epileptiform discharges with ictal

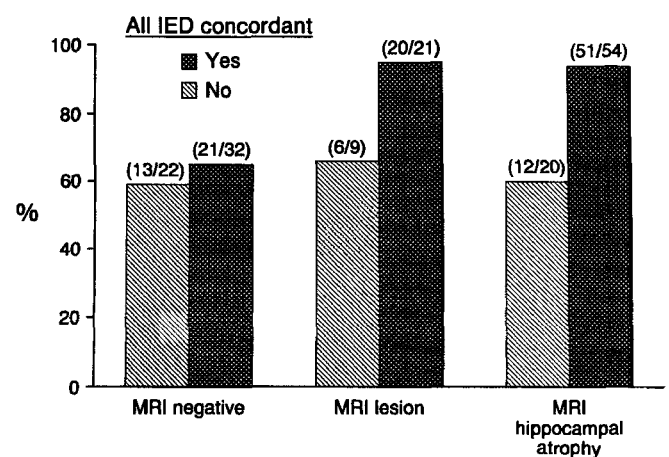


Figure. Comparison of the rates of excellent postsurgical outcome between patients with and without concordance of all interictal epileptiform discharges with the ictal focus. Differences between the two groups were significant ($p < 0.05$) when MRI detected unilateral hippocampal formation atrophy or other temporal lobe lesions. Numbers of patients are included in parentheses. IED = interictal epileptiform discharges.

onset in the temporal lobe was an important factor for excellent postsurgical outcome in patients with, but not in patients without, potentially epileptogenic abnormalities on MRI. Conversely, MRI abnormalities were an important factor for excellent outcome in patients with, but not in patients without, concordant interictal epileptiform discharges. In patients with concordant interictal epileptiform discharges, outcome was excellent in 95% when MRI abnormalities were present, but in only 65% when absent ($p = 0.001$). In contrast, for patients without concordant interictal epileptiform discharges, there was no difference if MRI abnormalities were present or absent (62% versus 59%; $p = 0.879$).

Discussion. Ever since ATL was proposed almost half a century ago as a specific treatment for medically unresponsive epilepsy,^{12,13} several studies have been conducted to identify factors that might be predictive of postsurgical outcome. Earlier attempts predated the standard use of statistics, and hence, only descriptive observations could be made.^{14,15} Bengzon et al.² first applied univariate statistics to identify factors associated with temporal lobectomy outcome. They reported that more than one-half the factors evaluated were of prognostic value, but altogether, 35 factors were analyzed without correction for multiple statistics. With 19 factors reportedly of prognostic significance, clinical application of the great number of factors becomes difficult because individual patients may have variable combinations of different factors. It should also be emphasized that the factors were evaluated by comparing two selected groups of patients representing the best and the worst postsurgical outcomes, whereas our study was a cohort study of consecutive ATL patients. Our assessment for prognostic factors also included modern presurgical tests, specifically MRI with hippocampal volume measurements and prolonged EEG recordings with video monitoring.

We limited our analyses to eight presurgical and six postsurgical factors, but correction for multiple statistics was not applied a priori to the univariate analysis. Nonetheless, the factors subjected to multivariate analysis were statistically significant on univariate analyses at very low p values (0.016 and less). Significance level was also set at $p < 0.05$ for factors to enter our logistic regression model, and all factors entered the model at $p = 0.024$ or less. In contrast, the only other study with multivariate analysis involving only ATL patients had set a higher significance level of $p < 0.10$.⁶

The presence of hippocampal formation atrophy on MRI is widely recognized to be associated with improvement in seizure control after ATL.^{16,17} The prognosis is also known to be favorable when interictal epileptiform discharges are restricted to the surgical focus.¹⁸⁻²⁰ However, until our current study, it was not known whether the prognostic value of well-localized interictal epileptiform discharges was independent of the presence of hippocampal formation atrophy. Our study reveals that the presence of scalp

interictal epileptiform discharges exclusively in the location of ictal onset is just as important as MRI detection of hippocampal formation atrophy. The presence of MRI-detected hippocampal formation atrophy alone does not confer the highest probability of excellent postsurgical outcome. Without interictal epileptiform discharges that are all concordant with ictal onset, only about 60% of ATL patients will have excellent postsurgical seizure control, whether hippocampal formation atrophy is present or absent on MRI. On the other hand, the presence of the two independently predictive factors of MRI-detected hippocampal formation atrophy and concordant interictal epileptiform discharges will identify patients with the highest probability of excellent outcome. Their exceptionally high chance of over 90% excellent outcome could be explained by studies showing that bilateral temporal lobe abnormalities are common in intractable epilepsy patients. In a review of autopsy series of temporal lobe epilepsy patients, Babb and Brown²¹ concluded that up to 90% of the patients had bitemporal damage. Their review surprisingly suggests that mesial temporal sclerosis is more often a bilateral disease. More recent investigations of living patients with magnetic resonance spectroscopy also reveals evidence of reduced intensity of the *N*-acetylaspartate signal in the temporal lobe contralateral to the apparent seizure focus.^{22,23} King et al.²⁴ emphasized that hippocampal formation atrophy is not an absolute indicator of ictal onset. Surface ictal onsets in 27% of their patients and depth ictal onsets in 10% were not concordant with the unilateral hippocampal formation atrophy. The foregoing studies support our finding that both neuroimaging and electrophysiologic abnormalities are needed to distinguish those whose epileptogenic focus is well-localized to one temporal lobe.

The importance of finding unilateral interictal epileptiform discharges was further demonstrated by Steinhoff et al.,²⁵ who found that patients with bitemporal interictal epileptiform discharges are more likely to have seizures arising independently from both temporal lobes. We have previously reported that the results of routine EEG are not associated with surgical outcome,²⁶ but the study relied on a single outpatient EEG recording. Furthermore, the study did not evaluate the concordance of interictal epileptiform discharge with ictal onset. Our current study was based on all outpatient EEGs and on long-term EEG monitoring. Increased sampling of the interictal activity, and the use of ictal discharge location to determine concordance, improved our assessment of the value of interictal activity at the seizure focus.

The past occurrence of febrile seizures has been reported to be a favorable prognostic factor.^{27,28} Although our current study, as well that of Duncan and Sagar,²⁹ failed to demonstrate the prognostic value of febrile seizure history, we did find in a subsequent study³⁰ that a history of prolonged febrile seizures (>15 minutes) increases the probability of

excellent postsurgical outcome in patients with non-lesional temporal lobe epilepsy (i.e., MRI-negative or showing hippocampal formation atrophy). This observation involving prolonged febrile seizures in non-lesional temporal lobe epilepsy was first made by Falconer and colleagues,^{31,32} who then speculated a causal relationship between prolonged febrile seizures and mesial temporal sclerosis.

Unlike presurgical EEG recordings, results of postresection electrocorticography in our study were not associated with surgical outcome. Our experience is supported by the study of Tran et al.,³³ who also found no association between residual spikes on postresection electrocorticography and postsurgical seizure control. Others have reported that postsurgical outcome in patients with postresection electrocorticography spikes is worse than in patients without.^{27,34,35} The lack of agreement of these recent studies may be due to differences in the recording techniques and in the anesthetic agents used during surgery.

We found on univariate analysis that the presence of spikes or sharp waves on scalp EEG performed after ATL is associated with a lower probability of excellent seizure control at last follow-up. Of three recent studies, results in two^{36,37} agree with our finding, but findings in the remaining study³⁸ did not. Marked discrepancy in study designs makes it difficult to compare these studies with our own. Two of the studies involved both temporal and extratemporal epilepsy surgery patients,^{36,37} one of which³⁷ included for evaluation any EEG performed between 6 and 18 postsurgical months. Our EEG assessment was at 3 months after surgery, and multivariate analysis shows that it does not independently predict surgical outcome. A better postsurgical predictor of outcome in our study is the degree of seizure control during the first year. Ninety-four percent of patients who were seizure free in the first postoperative year had excellent outcome at last follow-up. On the other hand, only 42% of those who were not seizure free in the first postoperative year eventually had excellent outcome. Engel et al.¹ also reported that seizure control in the first postoperative year is of prognostic value: 57% of their patients with rare seizures in the first year became seizure free for at least 2 years by the end of 5 years of follow-up compared with 38% of those who had seizures in the first postoperative year. A more recent report found that early postoperative seizure control during the first week also predicts favorable, although not necessarily seizure free, outcome at last follow-up.⁶ The study also stratified the analysis for predictive factors by duration of follow-up. We have shown in this study that postsurgical outcome was not associated with the duration of follow-up, and our previous study¹⁰ also revealed that postsurgical seizure frequency of patients in this cohort was stable from one year to the next. By a priori design, our current study was intended to identify factors that are predictive of the latest outcome after ATL.

Despite our finding that unilateral hippocampal formation atrophy and concordance of interictal epileptiform discharges with the ictal discharge focus are independent and strong predictive factors, the absence of either does not preclude the possibility of postsurgical improvement in seizure control. With only one factor present, close to two-thirds of our patients still had excellent seizure control after ATL, with most being seizure free.

Furthermore, two series^{39,40} involving small numbers of patients have shown that seizure-free outcome is still possible in patients with bilateral hippocampal formation atrophy.

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